Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The equation of the image of circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is

(A) $x^2 + y^2 + 32x - 4y + 235 = 0$

(B) $x^2 + y^2 + 32x + 4y - 235 = 0$

(C) $x^2 + y^2 + 32x - 4y - 235 = 0$

(D) $x^2 + y^2 + 32x + 4y + 235 = 0$

Q.2

The point on the line $x + y = 4$ which lie at a unit distance from the line $4x + 3y = 10$, are

(A) $(3, 1), (-7, 11)$

(B) $(3, 1), (7, 11)$

(C) $(-3, 1), (-7, 11)$

(D) $(1, 3), (-7, 11)$

Q.3

The reflection of the point $(4, -13)$ in the line $5x + y + 6 = 0$, is

(A) $(-1, -14)$

(B) $(3, 4)$

(C) $(1, 2)$

(D) $(-4, 13)$

Q.4

P is a point on either of the two lines $\mathrm{y} - \sqrt{3}|\mathrm{x}| = 2$ at a distance of 5 units from their point of intersection. The coordinates of the foot of the perpendicular from P on the bisector of the angle between them are

(A) $\left(0, \frac{4+5\sqrt{3}}{2}\right)$ or $\left(0, \frac{4-5\sqrt{3}}{2}\right)$ depending on which the point P is taken

(B) $\left(0, \frac{4+5\sqrt{3}}{2}\right)$

@IITJEE_Advanced

(C) $\left(0, \frac{4-5\sqrt{3}}{2}\right)$

(D) $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$

Q.5

If the image of point $\mathrm{P}(2,3)$ in a line L is $\mathrm{Q}(4,5)$, then the image of point $\mathrm{R}(0,0)$ in the same line is:

(A) $(2,2)$

(B) $(4,5)$

(C) $(3,4)$

(D) $(7,7)$

Q.6

The equation of a straight line, which passes through the point $(a,0)$ and whose perpendicular distance from the point $(2a,2a)$ is a, is

(A) $3x-4y-3a=0$

(B) $x-a=0$

(C) both (a) and (b)

(D) Neither (a) nor (b)

Q.7

If $2p$ is the length of perpendicular from the origin to the lines $\frac{x}{a}+\frac{y}{b}=1$, then $a^2, 8p^2, b^2$ are in

(A) A.P.

(B) GP.

(C) H.P.

(D) None of these

Q.8

If $\mathrm{p}_1, \mathrm{p}_2$ are the lengths of the normals drawn from the origin on the lines $x\cos\theta+y\sin\theta=2a\cos 4\theta$ and
$x\sec\theta+y\operatorname{cosec}\theta=4a\cos 2\theta$
respectively, and $mp_1^2+np_2^2=4a^2$. Then

(A) $\mathrm{m}=1, \mathrm{n}=1$

(B) $m=1, n=4$

(C) $\mathrm{m}=4, \mathrm{n}=1$

@IITJEE_Advanced

(D) $\mathrm{m}=1, \mathrm{n}=-1$

Q.9

Two straight lines passing through the point $\mathrm{A}(3,2)$ cut the line $2\mathrm{y}=\mathrm{x}+3$ and x -axis perpendicularly at P and Q respectively. The equation of the line PQ is

(A) $7x+y-21=0$

(B) $x+7y+21=0$

(C) $2x+y-8=0$

(D) $x+2y+8=0$

Q.10

What is the acute angle between the lines represented by the equations $y-\sqrt{3}x-5=0$ and $\sqrt{3}\mathrm{y}-\mathrm{x}+6=0$?

(A) 30°

(B) 45°

(C) 60°

(D) 75°

Q.11

The lines $2x=3y=-z$ and $6x=-y=-4z$

(A) are perpendicular

(B) are parallel

(C) intersect at an angle 45°

(D) intersect at an angle 60°

Q.12

The number of equilateral triangles with $y=\sqrt{3}(x-1)+2$ and $y=-\sqrt{3}x$ as two of its sides is

(A) 0

(B) 1

(C) 2

(D) None of these

Q.13

@IITJEE_Advanced

The diagonals of the parallelogram whose sides are $lx+my+n=0, lx+my+n'=0, mx+ly+n=0$ and
$mx+ly+n'=0$ include an angle

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{2}$

(C) $\tan^{-1}\left(\frac{1^2-m^2}{1^2+m^2}\right)$

(D) $\tan^{-1}\left(\frac{2\,\mathrm{lm}}{1^2+\mathrm{m}^2}\right)$

Q.14

What is the angle between the lines $x+y=1$ and $x-y=1$?

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Q.15

A ray of light passing through a point $(1,2)$ is reflected on the x -axis at point Q and passes through the point $(5,8)$. Then the abscissa of the point Q is

(A) -3

(B) $9/5$

(C) $13/5$

(D) None of these

Q.16

The axes being inclined at an angle of 60°, the angle between the two straight lines
$y=2x+5$ and $2y+x+7=0$ is

(A) 90°

(B) $\tan^{-1}(5/3)$

(C) $\tan^{-1}(\sqrt{3}/2)$

(D) $\tan^{-1}(5/\sqrt{3})$

Q.17

A straight line L through the point $(3,-2)$ is inclined at an

@IITJEE_Advanced

angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x -axis, then the equation of L is

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$

(B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$

(D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

Q.18

The range of value of α such that $(0, \alpha)$ lies on or inside the triangle formed by the lines $y + 3x + 2 = 0$, $3y - 2x - 5 = 0, 4y + x - 14 = 0$ is

(A) $5 < \alpha \leq 7$

(B) $\frac{1}{2} \leq \alpha \leq 1$

(C) $\frac{5}{3} \leq \alpha \leq \frac{7}{2}$

(D) None of these

Q.19

$\mathrm{P(m, n)}($ where $\mathrm{m, n}$ are natural numbers) is any point in the interior of the quadrilateral formed by the pair of lines $xy = 0$ and the two lines $2x + y - 2 = 0$ and $\mathrm{4x + 5y = 20}$. The possible number of positions of the point P is

(A) six

(B) five

(C) four

(D) eleven

Q.20

Let O be the orgin. If $\mathrm{A}(1, 0)$ and $B(0, 1)$ and $P(x, y)$ are points such that $xy > 0$ and $x + y < 1$, then

(A) P lies either inside the triangle OAB or in the third quadrant

(B) P cannot lie inside the triangle OAB

(C) P lies inside the triangle OAB

(D) P lies in the first quadrant only

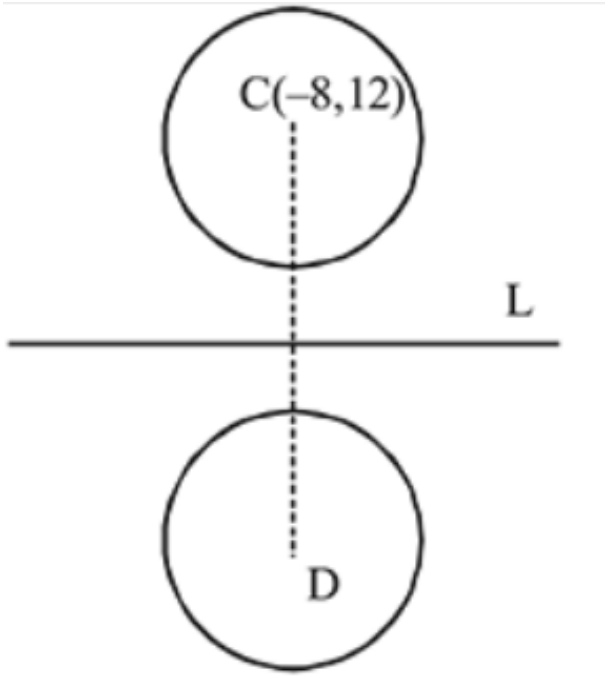
Answers & Solutions

Q.1 Answer:

@IITJEE_Advanced

$x^2 + y^2 + 32x + 4y + 235 = 0$

Solution:



The centre of the given circle is $(-8, 12)$ and radius is 5 . The image of the circle will have the same radius, i.e. the radius of the required circle is 5 . The centre D of the required circle is the image of the centre C of the given circle in the line mirror. If D be (α, β) then $\frac{\alpha+8}{4} = \frac{\beta-12}{7} = -2\left[\frac{4\times-8+7\times12+13}{4^2+7^2}\right]$ [See straight line]

or, $\frac{\alpha+8}{4} = \frac{\beta-12}{7} = \frac{-2\times65}{65} = -2$

$\therefore \alpha = -16, \beta = -2$

$\therefore$ **Required circle is** $(x+16)^2 + (y+2)^2 = 5^2$

Q.2 Answer:

$(3, 1), (-7, 11)$

Solution:

Let the point (h, k) lie on a line $x + y = 4$ then $h + k = 4$

and $1 = \pm\frac{4h+3k-10}{\sqrt{4^2+3^2}} \Rightarrow 4h + 3k = 15$

and $4h + 3k = 5$

On solving (i) and (ii); and (i) and (iii), we get the required points $(3, 1)$ and $(-7, 11)$. Trick : Check with options. Obviously, points $(3, 1)$ and $(-7, 11)$ lie on $x + y = 4$ and perpendicular distance of these points from $4x + 3y = 10$ is 1 .

Q.3 Answer:

$(-1, -14)$

Solution:

Let $\mathrm{Q}(a, b)$ be the reflection of $\mathrm{P}(4, -13)$ in the line $5x + y + 6 = 0$

Then the mid-point $R\left(\frac{a+4}{2}, \frac{b-13}{2}\right)$ lies on $5x + y + 6 = 0$

@IITJEE_Advanced

$\therefore 5\left(\frac{a+4}{2}\right)+\frac{b-13}{2}+6=0 \Rightarrow 5a+b+19=0 \ldots$ **i)**

Also PQ **is perpendicular to** $5x+y+6=0$ **Therefore** $\frac{b+13}{a-4} \times \left(-\frac{5}{1}\right)=-1 \Rightarrow a-5\,\mathrm{b}-69=0$.. **(ii)**

Solving (i) and (ii) we get $a=-1 \quad b=-14$

Q.4 Answer:

$\left(0, \frac{4+5\sqrt{3}}{2}\right)$

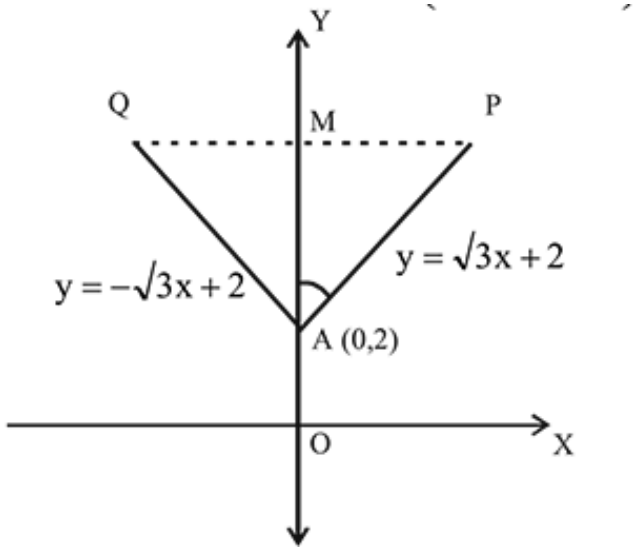
Solution:

Since $|x|=\begin{cases} x, x \geq 0 \\ -x, x<0 \end{cases}$

therefore the equations of two lines are $y=\sqrt{3}x+2, x \geq 0$ **and** $y=-\sqrt{3}x+2, x<0$

Clearly y-axis the only bisector of the angle between these two lines. There are two points P **and** Q **on these lines at a distance of 5 units from A. Clearly** M **is the foot of the perpendicular from** P **and** Q **on y -axis (bisector).** $\mathrm{AM}=\mathrm{AP}\cos 30^{\circ}=\frac{5\sqrt{3}}{2}$

Hence, coordinates of M **are**



Q.5 Answer:

$(7,7)$

Solution:

Mid-point of $P(2,3)$ **and** $Q(4,5)=(3,4)$ **Slope of** $\mathrm{PQ}=1$**, Slope of the line** $\mathrm{L}=-1$ **Mid-point** $(3,4)$ **lies on the line** L**. Equation of line** L**,** $y-4=-1(x-3) \Rightarrow x+y-7=0$ **...(i)**

Let image of point $\mathrm{R}(0,0)$ **be** $\mathrm{S}\left(x_1, y_1\right)$ **Mid-point of** $\mathrm{RS}=\left(\frac{x_1}{2}, \frac{y_1}{2}\right)$

Mid-point $\left(\frac{x_1}{2}, \frac{y_1}{2}\right)$ **lies on the line (i)** $\therefore x_1+y_1=14$ **...(ii)**

Slope of $\mathrm{RS}=\frac{y_1}{x_1}$**; Since** $\mathrm{RS} \perp$ **line** L

$\therefore \frac{y_1}{x_1} \times(-1)=-1 \quad \therefore x_1=y_1$ **...(iii)**

From (ii) and (iii), $x_1=y_1=7$ **Hence the image of** $\mathrm{R}=(7,7)$

Q.6 Answer:

both (a) and (b)

@IITJEE_Advanced

Solution:

Equation of line passing through $(\mathrm{a}, 0)$ **is** $\mathrm{y}=\mathrm{m}(\mathrm{x}-\mathrm{a}) \Rightarrow mx-y-ma=0$

Its distance from the point $(2\mathrm{a}, 2\mathrm{a})$ **is** $\left|\frac{2\mathrm{am}-2\mathrm{a}-\mathrm{ma}}{\sqrt{\mathrm{m}^2+1}}\right|=\mathrm{a}$ **(given)**

$\Rightarrow (m-2)^2=\left(m^2+1\right) \Rightarrow m^2-4m+4=m^2+1$

$\Rightarrow 0\,\mathrm{m}^2-4\,\mathrm{m}+3=0 \Rightarrow \mathrm{m}=\frac{3}{4}, \infty$

The required equation of lines are, from (i) $3x-4y-3a=0$ **and** $x-a=0$

Q.7 Answer:

H.P.

Solution:

We have $2p=\left|\frac{0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}\right| \Rightarrow \frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{4p^2}$

$\frac{1}{a^2}+\frac{1}{b^2}=\frac{2}{8p^2} \Rightarrow \frac{1}{a^2}, \frac{1}{8p^2}, \frac{1}{b^2}$ **are in A.P.**

$\Rightarrow a^2, 8p^2, b^2$ **are in H.P.**

Q.8 Answer:

$m=1, n=4$

Solution:

$\mathrm{p}_1^2=4\mathrm{a}^2 \cos^2 4\theta$

$\mathrm{p}_2^2=\frac{16\mathrm{a}^2 \cos^2 2\theta}{\sec^2\theta+\operatorname{cosec}^2\theta}=16\mathrm{a}^2 \cos^2 2\theta \cos^2\theta \sin^2\theta$

$=\mathrm{a}^2 \sin^2 4\theta$

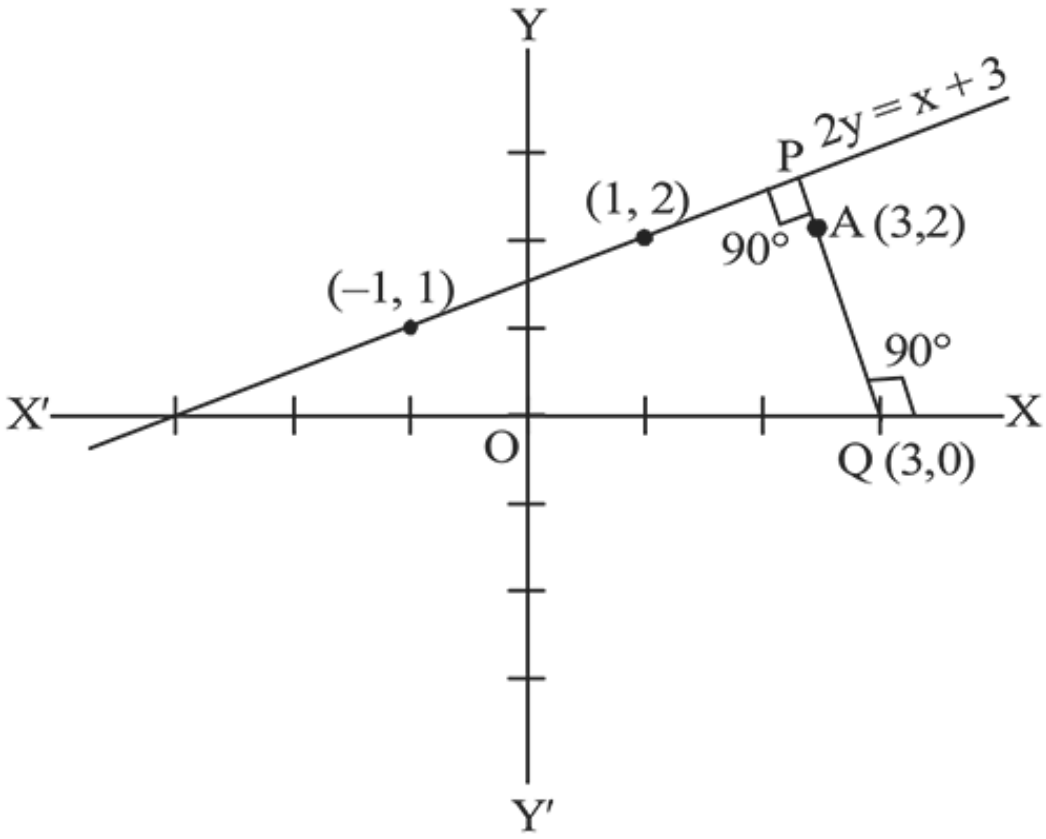
$\therefore \mathrm{p}_1^2+4\mathrm{p}_2^2=4\mathrm{a}^2$

Q.9 Answer:

$7x+y-21=0$

Solution:

@IITJEE_Advanced



∵ **Coordinates of** Q **are** $(3,0)$ **& it passes through** PQ.

∴ **Putting the values of** $(\mathrm{x}=3)\&(\mathrm{y}=0)$ **in options we get:**

Equation of line $\mathrm{PQ}=7\mathrm{x}+\mathrm{y}-21=0$

Q.10 Answer:

30°

Solution:

$y-\sqrt{3}x-5=0$ **line one**

$\sqrt{3}y-x+6=0$ line two

$y=mx+c$

$y=\sqrt{3}x+5 \qquad y=\frac{x}{\sqrt{3}}-\frac{6}{\sqrt{3}}$

$m_1=\sqrt{3} \qquad m_2=\frac{1}{\sqrt{3}}$

Angle between two lines,

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

$$=\left|\frac{\sqrt{3}-\frac{1}{\sqrt{3}}}{1+\sqrt{3}\frac{1}{\sqrt{3}}}\right|=\frac{1}{\sqrt{3}}$$

$$=\tan 30^{\circ}$$

$$\therefore \theta=30^{\circ}$$

Q.11 Answer:

@IITJEE_Advanced

are perpendicular

Solution:

$2x = 3y = -z$

or $\frac{x}{3} = \frac{y}{2} = \frac{z}{-6}$

$6x = -y = -4z$

or $\frac{x}{2} = \frac{y}{-12} = \frac{z}{-3}$

$\cos\theta = \frac{x_1x_2+y_1y_2+z_1z_2}{\sqrt{x_1^2+x_2^2+x_3^2}\cdot\sqrt{y_1^2+y_2^2+y_3^2}}$

$= \frac{(6-24+18)}{\sqrt{(3)^2+(2)^2+(-6)^2}\cdot\sqrt{(2)^2+(-12)^2+(-3)^2}}$

$\cos\theta = 0$

$\theta = 90^\circ$

So lines are perpendicular

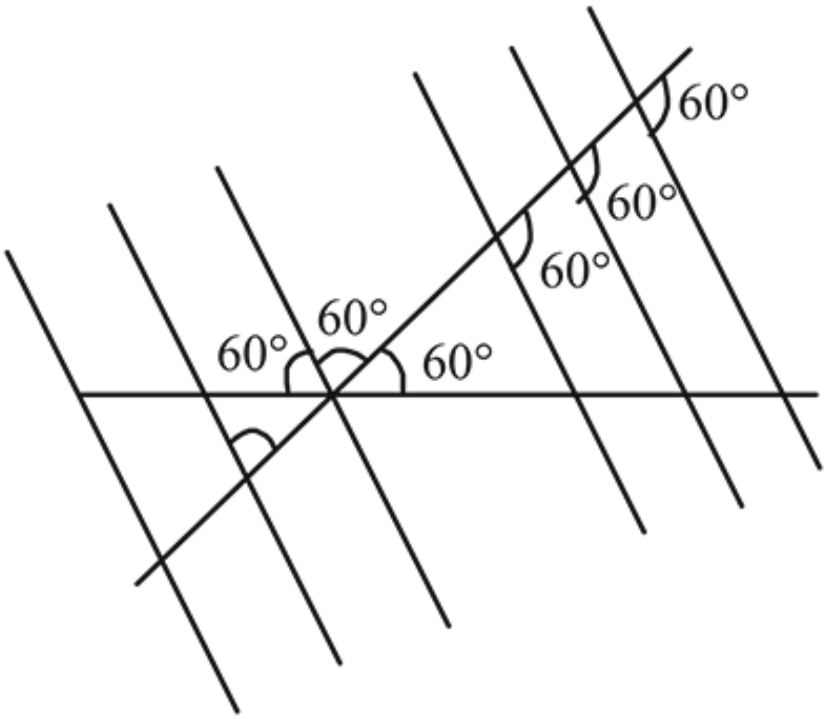
Q.12 Answer:

None of these

Solution:

The sides given are $y = \sqrt{3}(x-1)+2$ and $y = -\sqrt{3}x$. Lines are at an angle 60° to each other. Now any line parallel to obtuse angle bisector will make equilateral triangle with these lines as its two sides.

Therefore, infinitely many lines.



Q.13 Answer:

$\frac{\pi}{2}$

Solution:

(b) The perpendicular distance between the parallel lines in pairs is same, hence the lines form a rhombus

@IITJEE_Advanced

Q.14 Answer:

$\frac{\pi}{2}$

Solution:

Slope of $x+y=1$ **is** -1 **Slope of** $x-y=1$ **is 1** Let $\tan A=-1, \tan B=1$

$A=\frac{3\pi}{4}, B=\frac{\pi}{4}$

$A-B=\frac{\pi}{2}$

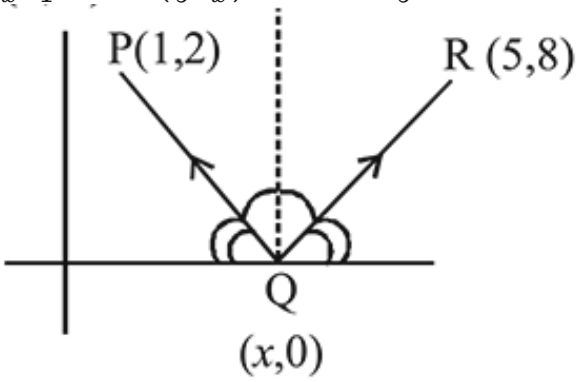
Q.15 Answer:

$9/5$

Solution:

Let point Q **be** $(x,0)$.

$\frac{0-2}{x-1}=-\left(\frac{8-0}{5-x}\right) \Rightarrow x=\frac{9}{5}$



Q.16 Answer:

$\tan^{-1}(5/\sqrt{3})$

Solution:

$\omega=60^{\circ}, \mathrm{m}_1=2,\ \mathrm{m}_2=-\frac{1}{2}$

$\tan\theta_1=\frac{m_1\sin\omega}{1+m_1\cos\omega}=\frac{2\times\sqrt{3}/2}{1+2\times1/2}=\frac{\sqrt{3}}{2}$

$\tan\theta_2=\frac{-1/2\times\sqrt{3}/2}{1-1/2\times1/2}=\frac{-\sqrt{3}}{4}\times\frac{4}{3}=\frac{-1}{\sqrt{3}}$

Let angle between the lines be f**. Then,**

$\tan\phi=\left|\frac{\tan\theta_1-\tan\theta_2}{1+\tan\theta_1\tan\theta_2}\right|=\left|\frac{\frac{\sqrt{3}}{2}+\frac{1}{\sqrt{3}}}{1-\frac{\sqrt{3}}{2}\times\frac{1}{\sqrt{3}}}\right|$

$\Rightarrow\phi=\tan^{-1}\left(\frac{5}{\sqrt{3}}\right)$

Q.17 Answer:

$y-\sqrt{3}x+2+3\sqrt{3}=0$

Solution:

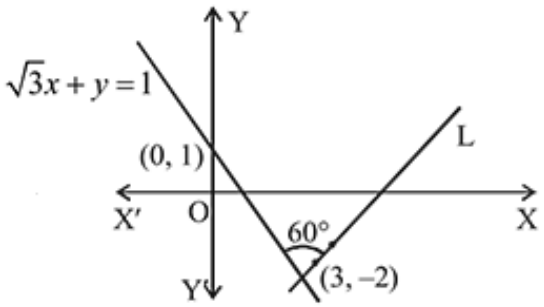
Let the slope of line L **be** m.

$\Rightarrow m=0$ **or** $m=\sqrt{3}$

$\because$ **L intersects** x **-axis,** $\quad\therefore m=\sqrt{3}$

@IITJEE_Advanced

$\therefore$ **Equation of L is $y+2=\sqrt{3}(x-3)$ or $\sqrt{3}x-y-(2+3\sqrt{3})=0$**



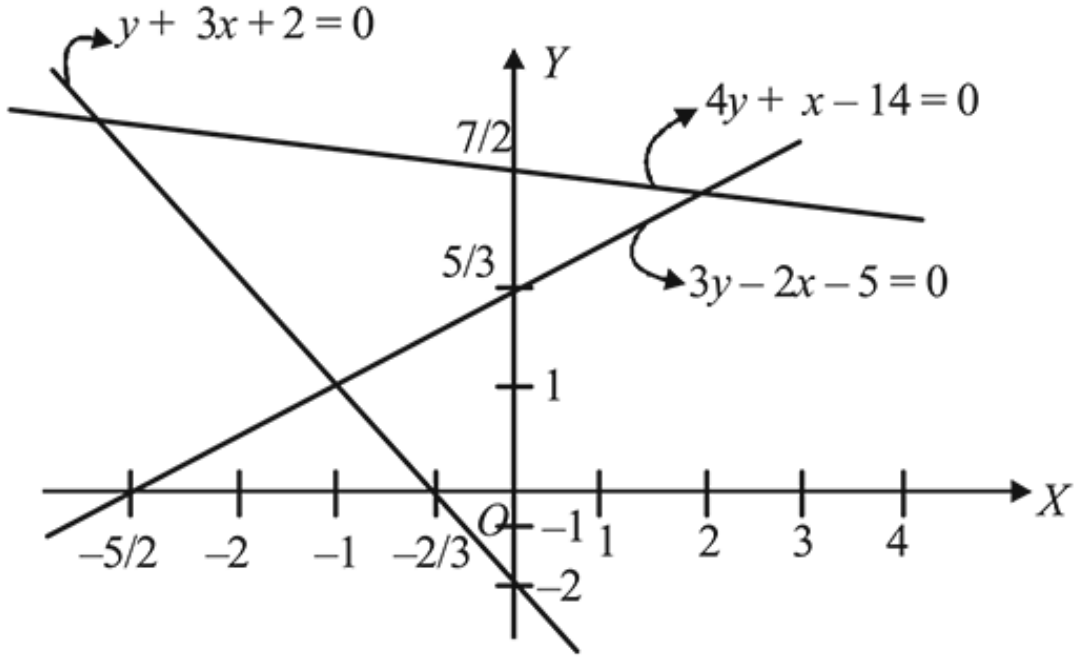
Q.18 Answer:

$\frac{5}{3} \leq \alpha \leq \frac{7}{2}$

Solution:

Point $(0, \alpha)$ lies on the y -axis. In such questions, it is easier to see the location of point if lines are drawn on the axis.

From the diagram, it is clear that $\frac{5}{3} \leq \alpha \leq \frac{7}{2}$

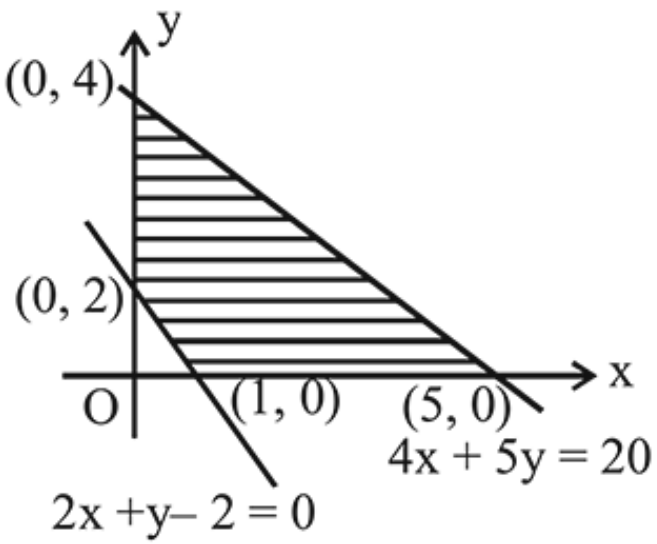


Q.19 Answer:

six

Solution:

@IITJEE_Advanced



It is clear from the figure $\mathrm{m} < 5$ **and** $\mathrm{n} < 4$**so,** $1 \leq m < 5$ **and** $2 \leq n < 4$ **Also** $2m + n - 2 > 0$ **and** $4\ \mathrm{m} + 5\mathrm{n} - 20 < 0$**. Under these conditons possible coordinates are** $(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 2)$

Q.20 Answer:

P lies either inside the triangle OAB or in the third quadrant

Solution:

Since $xy > 0, P$ **lies either in the first quadrant or in the third quadrant. The inequality** $x + y < 1$ **represents all points below the line** $x + y = 1$ **so that** $xy > 0$ **and** $x + y < 1$ **imply that** $\mathbf{P}$ **lies either inside the triangle OAB or in third quadrant.**

@IITJEE_Advanced